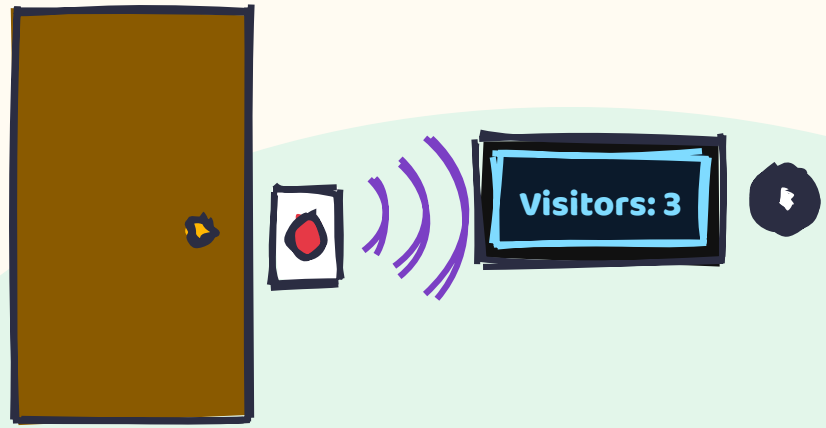
Young Innovators · Class 10

Wireless Doorbell with Visitor Counter

Two Arduinos talking by radio!

NRF24L01 radio | Addresses | Two-device system



Did you know?
Our radios use 2.4 GHz, the same band as Wi-Fi, Bluetooth and even microwave ovens.

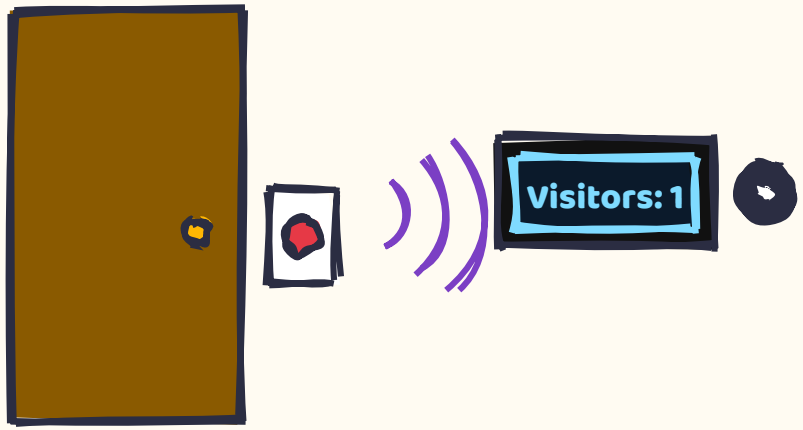
LEARN

Let's think...

How does the bell know?

The wire problem

An old doorbell needs a **wire** from the gate to inside the house. Drilling through walls is messy and expensive. Long wires break. Wireless doorbells need **no wire at all**.



A very long wire?

Shout louder?

Send a radio message!

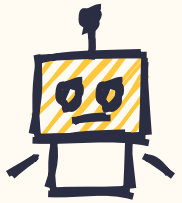
Today: pairs of groups build a **transmitter** for the door and a **receiver** that rings and counts visitors.

Learning aims

Today I will learn to…

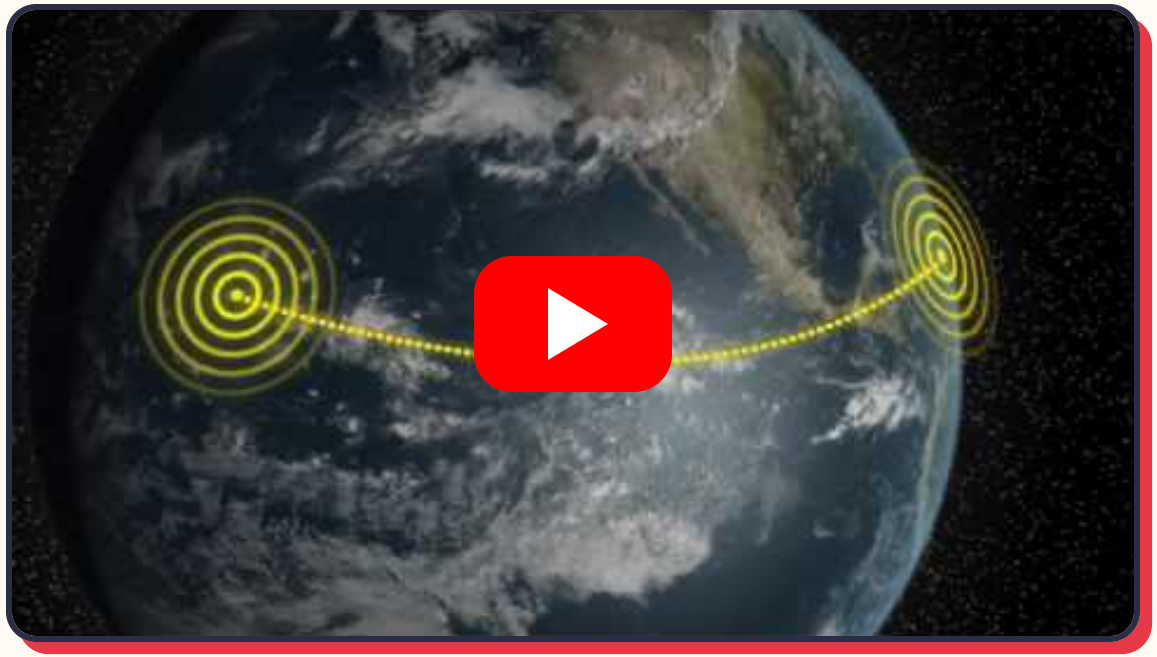
1. Explain how **radio waves** carry messages
2. Wire an **NRF24L01** safely to an Uno (3.3 V!)
3. Use matching **addresses** so the right devices talk
4. Program a **transmitter** and a **receiver** that rings and counts
5. Test the **range** through walls and explain the results

Keywords
radio · 2.4 GHz · transmitter · receiver · address · pipe · ACK



Watch and think

Video time!



Tour of the Electromagnetic Spectrum: Radio Waves

NASA Science · about 4 minutes · youtu.be/OzDmEA8x0nQ

Watch for...

1. Where do **radio waves** sit in the electromagnetic spectrum?
2. Name two things that use radio waves.
3. Can radio waves travel through walls?

Scan to watch

After the video: Our wireless doorbell uses radio. How far do you think it will reach through walls? Predict, then test!

The science

Radio waves: light we cannot see



2.4 GHz

Our radio wave wiggles **2.4 billion times every second**. It travels at the speed of light, so the message reaches the receiver almost instantly.

Walls and water

Radio passes through wood and bricks, but gets weaker. **Metal** and **water** (even people!) block it more.

Crowded band

Many devices share 2.4 GHz. Good radios use addresses and channels to avoid mix-ups.

Real world

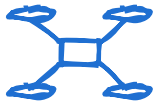
Short-range radio is everywhere



Wireless bells
Button outside, chime inside: exactly our project



Car keys
Press the fob, a coded radio message unlocks the car



Drone remotes
Send joystick moves many times a second



Walkie-talkies
Send voices on a shared radio channel

? Your car key and your neighbour's look the same. Why doesn't yours open their car?

Hint
Each key sends a different CODE. Our radios use ADDRESSES.

Meet the radio

The NRF24L01 module



VCC	3.3V (NOT 5V!)
GND	GND
CE / CSN	9 / 10
SCK	13
MOSI / MISO	11 / 12
IRQ	not used

3.3 V only!

Connecting VCC to 5V can **destroy** the module. Double-check before plugging in USB.

Add a capacitor

A **10 µF** capacitor across VCC and GND (long leg to VCC) smooths the power. Without it, messages often get lost.

SPI

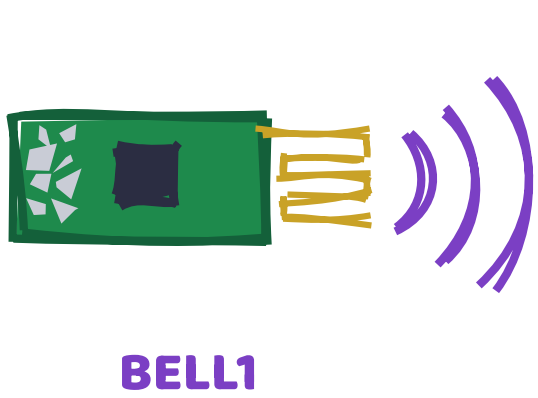
The module talks to the Uno through **SPI**, a fast 4-wire connection (SCK, MOSI, MISO, CSN). Pins 11, 12, 13 are fixed for SPI on the Uno.

Pin diagram

The pins are not labelled on most boards! Use the diagram your teacher gives you, looking from the top.

Big idea

Addresses: a name on the envelope







Matching address

The transmitter writes to address **"BELL1"**. Only a receiver listening on **"BELL1"** accepts the message. Up to 5 letters.

Classroom rule!

10 groups in one room = 10 doorbells. Every pair must pick its **own** address (BELL1, BELL2, …), or everyone's chime rings!

How they talk

Send, then wait for a 'got it!'



Who does what?

Door unit: button + transmitter.
Inside unit: receiver + buzzer + OLED counter.

Automatic acknowledgement

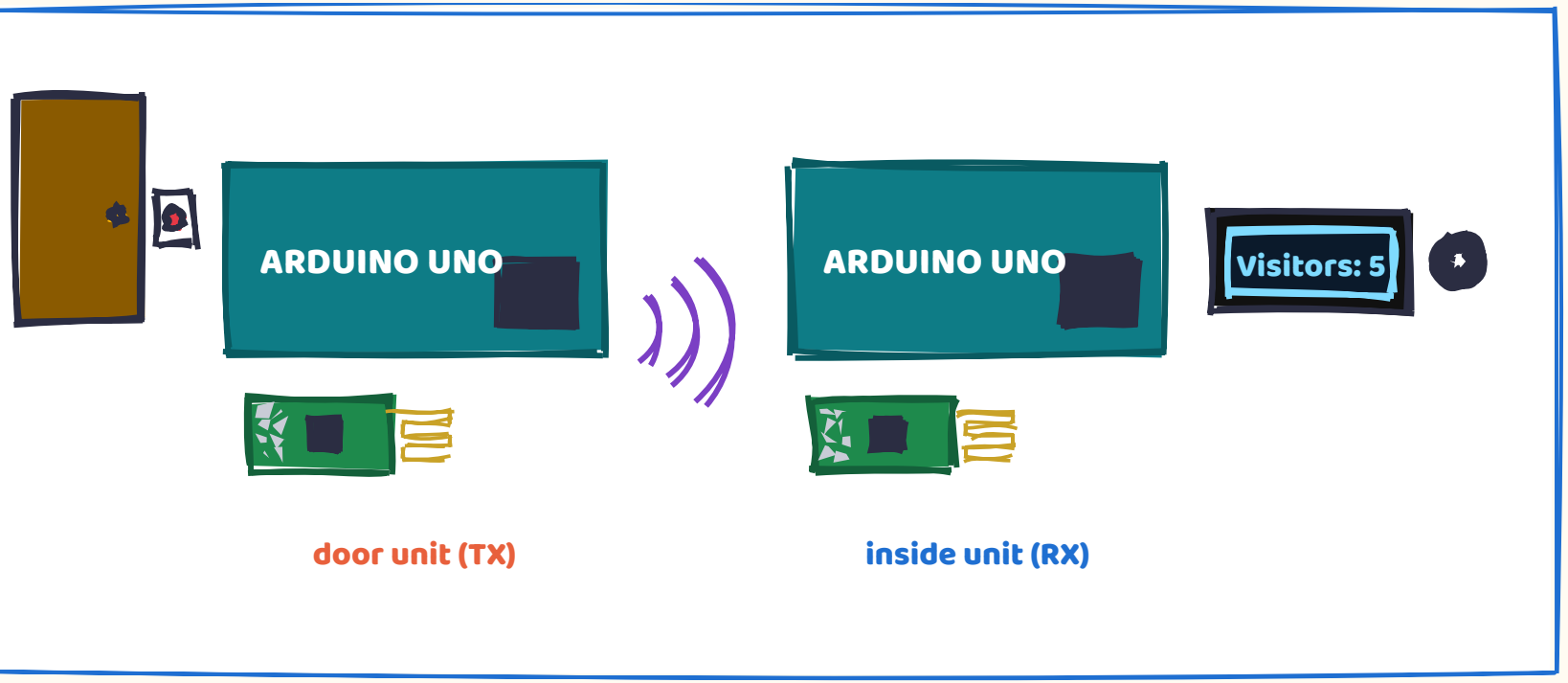
The NRF24 sends back a tiny **ACK** message by itself. **radio.write()** returns **true** if the ACK arrived. If not, it tries again a few times.

Use the ACK!

Light an LED on the door unit when the ACK arrives, so the visitor knows the bell rang.

The plan

Two Arduinos, one system



Pair up!

Two groups join together. One builds the **door unit**, one builds the **inside unit**. You must agree the address and the message.

System thinking

If one half fails, the whole system fails. Test each half on its own first!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Radio post office
2. Our doorbell kit (per pair of groups)
3. Wiring both units
4. The transmitter
5. The receiver and counter
6. How far can the bell reach?
7. Debug like a radio engineer

Class game

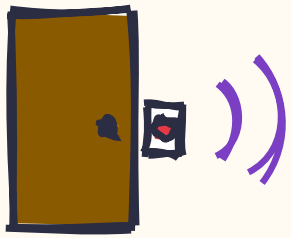
Radio post office

How to play

1. Each pair writes an **address** on a card (BELL1, BELL2…).
2. Senders write a message and the address on a paper slip.
3. The "radio" (teacher) shouts every message out loud to the whole room.
4. Only the pair with the matching address may reply "ACK!"
5. Then: two senders shout at the same time. What happens?

Then discuss

Why did shouting at the same time cause problems? How could radios take turns?



Big idea

Everyone hears every radio message. Addresses decide who listens.

Build it

Our doorbell kit (per pair of groups)

2 × Arduino Uno
TX and RX

2 × NRF24L01
The radios

2 × 10 µF capacitor
Power smoothing

Push button + 10k
The door button

OLED 0.96 in
Visitor counter

Buzzer
The chime

Also: 2 breadboards, jumper wires (M-F for the radios), 2 USB cables, 2 laptops, and the RF24 library by TMRh20.

Power for range tests
Use a power bank on the door unit so you can walk around.

MAKE

Build it

Wiring both units

NRF24L01 (both units)	Uno
VCC	3.3V
GND	GND
CE / CSN	9 / 10
SCK / MOSI / MISO	13 / 11 / 12
10 µF capacitor	across VCC and GND

Door unit	
Button	5V → pin 2
10k	pin 2 → GND

Inside unit	
OLED SDA / SCL	A4 / A5
Buzzer +	pin 8

Before you plug in USB

Ask a partner to check: VCC on **3.3V**? Capacitor the right way round? Then the teacher checks too.

MAKE

Code it: door unit

The transmitter

```
#include <SPI.h>
#include <RF24.h>
RF24 radio(9, 10);              // CE, CSN
const byte address[6] = "BELL1"; // your pair's address
const int BUTTON = 2;

void setup() {
  pinMode(BUTTON, INPUT);
  radio.begin();
  radio.openWritingPipe(address);
  radio.setPALevel(RF24_PA_LOW);
  radio.stopListening();         // I am the sender
}

void loop() {
  if (digitalRead(BUTTON) == HIGH) {
    const char msg[] = "RING";
    bool ok = radio.write(&msg, sizeof(msg));
    delay(1000);                 // one ring per press
  }
}
```

Key lines

openWritingPipe: who I send to.

stopListening: I am a sender.

write: send the bytes, returns true if ACK came back.

Why delay(1000)?

Holding the button for half a second would send many RINGs. The delay means one press = one visitor.

MAKE

Code it: inside unit

The receiver and counter

```
RF24 radio(9, 10);
const byte address[6] = "BELL1"; // must match!
int visitors = 0;

void setup() {
  radio.begin();
  radio.openReadingPipe(0, address);
  radio.setPALevel(RF24_PA_LOW);
  radio.startListening();        // I am the receiver
  // + OLED setup
}

void loop() {
  if (radio.available()) {
    char msg[32] = "";
    radio.read(&msg, sizeof(msg));
    if (strcmp(msg, "RING") == 0) {
      visitors++;
      tone(8, 880, 300); delay(350);
      tone(8, 660, 400);         // ding-dong!
      // show visitors on the OLED
    }
  }
}
```

Key lines

openReadingPipe: listen on my address.

available: has a message arrived?

strcmp: is it exactly "RING"? (0 means equal)

Why check the message?

Later you can send other messages ("KNOCK", "PARCEL") and react to each one differently.

Experiment

How far can the bell reach?

Door unit position	PA_LOW rings?	PA_MAX rings?
Same table (1 m)	yes / no	yes / no
Across the room (8 m)	yes / no	yes / no
Through 1 wall	yes / no	yes / no
Down the corridor (20 m)	yes / no	yes / no
Through 2 walls	yes / no	yes / no

Fair test

Keep the inside unit in one place. Press 5 times at each spot. Ring 5 out of 5 = reliable.

PA levels

PA_MIN, LOW, HIGH, MAX: more power = more range, but more battery used.

Explain your results

Where did it stop working? Was it distance or the walls? What else was in the way: metal doors, people, water pipes?

Test like an engineer

Debug like a radio engineer

Problem	Try this
Nothing ever arrives	VCC on 3.3V? Capacitor fitted?
Works sometimes only	add/replace the capacitor, shorter wires
Another group's bell rings	addresses are the same: change yours
Upload error	check the RF24 library is installed
Rings many times per press	add the delay, or detect the button press edge

Debug checklist

Print **ok** from radio.write() to the Serial monitor. true = the ACK came back. Test both units on one table first.

Swap ONE part at a time with a working pair (radio, then wires, then Arduino) to find the faulty part.

MAKE

Level up!

Doorbell challenges

Choose your challenge

Visit times: send millis() and show "minutes since switch-on" for the last visitor.

Two doors: add a back-door button that sends a different message and a different tune.

Do not disturb: a button on the inside unit that silences the bell but still counts.

Reply: make the inside unit send "COMING" back, and light an LED at the door.

Think like an engineer

Anyone with an NRF24 who knows your address could ring your bell. How could you make it more secure?

Boss level

Send a secret code number with each RING, and only ring if the code is right.

Show what you know

What did we learn?

- ☑ Radio waves carry messages without wires
- ☑ The NRF24L01 needs 3.3 V and a capacitor
- ☑ Matching addresses decide who listens
- ☑ The ACK tells the sender the message arrived
- ☑ Walls, metal and distance reduce range

At home

Find 3 wireless gadgets at home. Do they use Wi-Fi, Bluetooth or another radio?



? **Exit ticket:** why must every pair in the class use a different address?

Next class: a Wi-Fi Automatic Plant Watering System!